

SHUBHAM PATIL

AI Engineer | Agentic Systems & LLM Applications

+91 9373868631 • sup31patil@gmail.com • Sangli, Maharashtra • LinkedIn • GitHub

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building multi-agent orchestration systems, RAG pipelines, MCP servers, and tool-calling integrations. Deploys production-ready LLM applications via FastAPI and Docker, and integrates vector databases with Hugging Face models. National hackathon finalist (EY Techathon 6.0) recognized for Generative AI excellence.

TECHNICAL SKILLS

Agentic AI & Orchestration: CrewAI, LangGraph, LangChain, Multi-Agent Systems, MCP (Model Context Protocol), Tool/Function Calling

LLMs & APIs: Gemini Pro, Groq API, OpenAI API, Ollama, Prompt Engineering, RAG Pipelines

Vector DBs & Embeddings: ChromaDB, FAISS, Hugging Face Embeddings, Sentence Transformers, Semantic Search, Document Chunking

Backend & Deployment: FastAPI, Uvicorn, Docker, REST APIs, Render, Hugging Face Spaces, Vercel, GitHub Actions

Full-Stack: MERN (MongoDB, Express.js, React, Node.js), Python

EXPERIENCE

AI Engineer — Hackathon & Project Experience — National-Level Competitions 2026 – Present

Built and deployed end-to-end AI systems across 10+ national hackathons using MCP, Tool Calling, CrewAI, ChromaDB, Hugging Face, and FastAPI.

- **EY Techathon 6.0 — National Finalist:** PharmaFlow-AI, a multi-agent pharmaceutical research platform built with CrewAI, Groq API, and a Dockerized FastAPI deployment.
- **ET Hackathon — Semi-Finalist:** AI data synthesis solution using Hugging Face Transformers, LangChain RAG, and tool calling for external API orchestration, deployed via Dockerized FastAPI.
- **Cognizant & 8+ National Events:** Delivered agentic prototypes — semantic search engines, MCP server integrations, LLM tool-calling pipelines, and multi-agent automation — within 24–48 hr deadlines.

TECHNICAL PROJECTS

PharmaFlow-AI — AI Engineer (Agentic Systems) 02/2026

Python • FastAPI • React • Gemini Pro • Groq API • Tool Calling • MongoDB • Docker

- Automated **85%** of manual pharma research processes via asynchronous multi-agent pipelines with real-time REST API orchestration.
- Integrated 5+ biomedical databases (PubChem, ChEMBL, ClinicalTrials.gov, OpenTargets) featuring hybrid Gemini/Groq LLM failover for sub-500ms synthesis.
- Deployed a Dockerized FastAPI backend, MongoDB database layer, and a React frontend hosted on Vercel.
- Delivered a 40% improvement in research output via an automated clinical, patent, and market report generation engine.

Chat with Research Paper Assistant — AI Engineer 05/2026

Python • LangChain • ChromaDB • Hugging Face • FastAPI • React • Streamlit • Tool Calling

- Built a full RAG pipeline: LangChain + ChromaDB + Hugging Face sentence-transformer embeddings, PDF chunking, and semantic indexing with tool calling for dynamic context retrieval.
- Developed context-aware Q&A and paper summarization using LLM-powered answer generation over indexed research PDFs.
- Deployed a Streamlit app on Hugging Face Spaces with a FastAPI inference endpoint for programmatic access, significantly reducing research review time.

EDUCATION

B.Tech — Computer Science & Engineering — PVPIT, Sangli, Maharashtra 07/2024 – 07/2027

Diploma — Computer Engineering — PVPIT, Sangli, Maharashtra 07/2021 – 07/2024

CERTIFICATIONS & ACHIEVEMENTS

- **AI Engineer Agentic Track:** The Complete Agent & MCP Course — Lingency (09/2025)
- **National Finalist | EY Techathon 6.0:** PharmaFlow-AI recognized among top national teams for Generative AI & agentic systems.
- **Semi-Finalist | ET Hackathon:** AI-driven solution using RAG, Tool Calling, and scalable deployment.
- **10+ Hackathons — Cognizant & National Events:** Consistently built and deployed AI/agentic systems under competitive deadlines.